

# Dr. Ryan Chown

Assistant Professor

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<https://rchwn.github.io>

## INTERESTS AND EXPERTISE

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The interstellar medium and star formation in galaxies. Application of mathematical and computational techniques to astronomical research and large datasets. Infrared and radio observations with JWST, ALMA and the VLA.

## EDUCATION

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**McMaster University** Hamilton, ON  
Ph.D. Astronomy 08/2021  
**Thesis:** *Multi-wavelength studies of the interstellar medium and star formation in nearby galaxies*  
**Advisors:** Prof. Christine D. Wilson and Prof. Laura Parker  
Finalist, J. S. Plaskett Medal for Most Outstanding PhD Thesis, Canada

**McGill University** Montreal, QC  
M.Sc. Physics 05/2015 - 05/2017  
**Thesis:** *Mapping the Millimeter-wave Sky with Combined South Pole Telescope and Planck Data*  
**Advisor:** Prof. Gil Holder

**McGill University** Montreal, QC  
B.Sc. Physics, with a Minor in Computer Science 09/2012 - 04/2015

## ACADEMIC POSITIONS

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**Algoma University** Sault Ste. Marie, ON  
Assistant Professor (Tenure-Track) 9/2025-  
**The Ohio State University** Columbus, OH  
Postdoctoral Scholar 10/2023-9/2025  
**The University of Western Ontario** London, ON  
Postdoctoral Scholar 10/2021-09/2023

## GRANTS

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NSERC Discovery Grant, \$195,000 CAD over 5 years 5/2026  
NSERC Discovery Launch Supplement, \$12,500 CAD 5/2026

## TEACHING EXPERIENCE

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PHYS 1906 Instructor Algoma  
*General Astronomy I* S26  
MATH 1036 Instructor Algoma  
*Calculus I* S26  
MATH 2056 Instructor Algoma

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|------------------------------------------------|------------------|
| <i>Discrete Mathematics II</i>                 | W26              |
| PHYS 1007 Instructor                           | Algoma           |
| <i>Introductory Physics II</i>                 | W26              |
| PHYS 1006 Instructor                           | Algoma           |
| <i>Introductory Physics I</i>                  | W25              |
| PHYS 1AA3 Teaching Assistant                   | McMaster         |
| <i>Introduction to Modern Physics</i>          | 2020             |
| PHYS 1A03 Teaching Assistant                   | McMaster         |
| <i>Introductory Physics</i>                    | 2018, 2019, 2020 |
| PHYS 1E03 Teaching Assistant                   | McMaster         |
| <i>Waves, Electricity, and Magnetic Fields</i> | 2018             |
| ASTRO 1F03 Teaching Assistant                  | McMaster         |
| <i>Astronomy and Astrophysics</i>              | 2017, 2021       |
| ASTRO 2C03 Teaching Assistant                  | McMaster         |
| <i>Big Questions</i>                           | 2017, 2021       |
| PHYS 183 Teaching Assistant                    | McGill           |
| <i>The Milky Way and Beyond</i>                | 2016 and 2017    |
| PHYS 230 Teaching Assistant                    | McGill           |
| <i>Dynamics of Simple Systems</i>              | 2015 and 2016    |

## RESEARCH SUPERVISION EXPERIENCE

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NOTES: \* indicates that I was the primary (informal) supervisor

† = student held a MITACS Globalink Research Award

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|----------------------------------------------------------------------------------------------|------|
| 5. *†Rajarshi Choudhury, <i>Indian Inst. of Sci. Ed. and Research Bhopal (undergraduate)</i> | 2023 |
| 4. †Arnab Chowhan, <i>Centre for Excellence in Basic Sciences, Mumbai (undergraduate)</i>    | 2023 |
| 3. *†Raphaella Kestler, <i>Durham University (undergraduate)</i>                             | 2022 |
| 2. *†Holly Raynor, <i>Durham University (undergraduate)</i>                                  | 2022 |
| 1. *Khaleda Ramzi, <i>Western University Work-Study Program (undergraduate)</i>              | 2022 |

## SERVICE

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|                                              |       |
|----------------------------------------------|-------|
| External Panelist for JWST Cycle 5 Proposals | 2025  |
| Referee for ApJ, A&A, MNRAS                  | 2023– |

## PROFESSIONAL DEVELOPMENT

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|                                                                                                                                                                                                     |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Western Certificate for University Teaching and Learning                                                                                                                                            | January 2024 |
| - An intensive five-component program designed “to enhance the quality of teaching by graduate students and postdoctoral scholars, and to prepare them for a future faculty or professional career” |              |
| - <a href="https://teaching.uwo.ca/programs/certificates/cut1.html">https://teaching.uwo.ca/programs/certificates/cut1.html</a>                                                                     |              |
| “What makes a great postdoc mentor” – Workshop at UWO                                                                                                                                               | January 2022 |

## PUBLICATIONS

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*Overall summary: 87 peer-reviewed papers; 2393 citations; h-index=30*

**A. ≤ 3rd author papers**

*Summary: 11 papers, 382 citations as of May , 2026.*

11. Sutter, J., Sandstrom, K., **Chown, R.**, et al., *Characterization of Two Cool Galaxy Outflow Candidates Using Mid-infrared Emission from Polycyclic Aromatic Hydrocarbons*. ApJL 992 (2025): L7
10. **Chown, R.**, et al., *Relationships between Polycyclic Aromatic Hydrocarbons, Small Dust Grains, H<sub>2</sub>, and H I in Local Group Dwarf Galaxies NGC 6822 and WLM Using JWST, ALMA, and the VLA*. ApJ 987 (2025): 91
9. **Chown, R.**, et al., *PDRs4All: XIII. Empirical prescriptions for the interpretation of JWST imaging observations of star-forming regions*. A&A 698 (2025): A86
8. **Chown, R.**, et al., *Polycyclic Aromatic Hydrocarbon and CO(21) Emission at 50150 pc Scales in 70 Nearby Galaxies*. ApJ 983 (2025): 64
7. **Chown, R.**, et al., *PDRs4All. IV. An embarrassment of riches: Aromatic infrared bands in the Orion Bar*. A&A 685 (2024): A75
6. **Chown, R.**, et al., *The cold gas and dust properties of red star-forming galaxies*. MNRAS 516 (2022): 84-99
5. **Chown, R.**, et al., *A new estimator of resolved molecular gas in nearby galaxies*. MNRAS 500 (2021): 1261-1278
4. **Chown, R.**, et al., *Linking bar- and interaction-driven molecular gas concentration with centrally enhanced star formation in EDGE-CALIFA galaxies*. MNRAS 484 (2019): 5192-5211
3. **Chown, R.**, et al., *Maps of the Southern Millimeter-wave Sky from Combined 2500 deg<sup>2</sup> SPT-SZ and Planck Temperature Data*. ApJS 239 (2018): 10
2. Omori, Y., **Chown, R.**, et al., *A 2500 deg<sup>2</sup> CMB Lensing Map from Combined South Pole Telescope and Planck Data*. ApJ 849 (2017): 124
1. Crawford, T., **Chown, R.**, et al., *Maps of the Magellanic Clouds from Combined South Pole Telescope and PLANCK Data*. ApJS 227 (2016): 23

## **B. Other papers with significant contributions during or after PhD**

*Summary: 59 papers, 1178 citations as of May , 2026.*

59. Hassani, H., et al., *The Hidden Life of Stars: Embedded Beginnings to Asymptotic Giant Branch Endings in the PHANGSJWST Sample. I. Catalog of Mid-infrared Sources*. ApJS 284 (2026): 3
58. Khan, B., et al., *PDRs4All: XX. The 69  $\mu$ m region as a probe of PAH charge and size in the Orion Bar*. A&A 709 (2026): A39
57. Maragkoudakis, A., et al., *PDRs4All: XIX. The evolution of the PAH ionisation and PAH size distribution across the Orion Bar*. A&A 709 (2026): A38
56. Bazzi, Z., et al., *The lifetime of 100,000 molecular clouds in the nearby Universe*. arXiv e-prints (2026): arXiv:2604.24759
55. Kim, J., et al., *Localized Deviations from the CO/Polycyclic Aromatic Hydrocarbon Relation in PHANGS-JWST Galaxies: Faint Polycyclic Aromatic Hydrocarbon Emission or Elevated CO Emissivity?*. ApJ 1001 (2026): 67
54. Lopez, S., et al., *JWST Observations of Starbursts: Polycyclic Aromatic Hydrocarbons Closely Trace the Cool Phase of M82's Galactic Wind*. ApJL 999 (2026): L7

53. McClain, R., *et al.*, *Resolved H II Regions in NGC 253: Ionized Gas Structure and Suggestions of a Universal Density Surface Brightness Relation*. ApJ 998 (2026): 166
52. Ramambason, L., *et al.*, *Duration and properties of the embedded phase of star formation in 37 nearby galaxies from PHANGS-JWST*. A&A 706 (2026): A186
51. Bazzi, Z., *et al.*, *PHANGS-JWST: The largest extragalactic molecular cloud catalog traced by polycyclic aromatic hydrocarbon emission*. A&A 706 (2026): A40
50. Zannese, M., *et al.*, *PDRs4All: XVII. Formation and excitation of HD in photodissociation regions: Application to the Orion Bar*. A&A 705 (2026): A128
49. Tarantino, E., *et al.*, *JWST Captures Growth of Aromatic Hydrocarbon Dust Particles in the Extremely Metal-poor Galaxy Sextans A*. arXiv e-prints (2025): arXiv:2512.04060
48. Sun, J., *et al.*, *Resolved Profiles of Stellar Mass, Star Formation Rate, and Predicted CO-to-H<sub>2</sub> Conversion Factor Across Thousands of Local Galaxies*. ApJ 994 (2025): 263
47. Graham, G., *et al.*, *PAH Marks the Spot: Digging for Buried Clusters in Nearby Star-forming Galaxies*. AJ 170 (2025): 340
46. Pathak, D., *et al.*, *Masses, Star Formation Efficiencies, and Dynamical Evolution of 18,000 H II Regions*. ApJL 993 (2025): L20
45. Oakes, E., *et al.*, *The Hierarchical Dynamical State of Molecular Gas from 3 to 300 pc in NGC 253*. ApJ 993 (2025): 193
44. Egorov, O., *et al.*, *Polycyclic aromatic hydrocarbon destruction in star-forming regions across 42 nearby galaxies*. A&A 703 (2025): A103
43. Koch, E., *et al.*, *The Karl G. Jansky Very Large Array Local Group L-Band Survey (LGLBS)*. ApJS 279 (2025): 35
42. Sarbadhicary, S., *et al.*, *A First Look at Spatially Resolved Infrared Supernova Remnants in M33 with JWST*. ApJ 989 (2025): 138
41. Kim, J., *et al.*, *Timescales of Polycyclic Aromatic Hydrocarbon and Dust Continuum Emission from Gas Clouds Compared to Molecular Gas Cloud Lifetimes in PHANGS-JWST Galaxies*. ApJ 988 (2025): 215
40. Khan, B., *et al.*, *PDRs4All: XIV. Probing CH out-of-plane bending modes of PAH molecules in the Orion Bar with JWST*. A&A 699 (2025): A133
39. Elmegreen, B., *et al.*, *An Investigation of Disk Thickness in M51 from H $\alpha$ , Pa $\alpha$ , and Mid-infrared Power Spectra*. ApJ 986 (2025): 13
38. Leroy, A., *et al.*, *Cloud-scale Gas Properties, Depletion Times, and Star Formation Efficiency per Freefall Time in PHANGSALMA*. ApJ 985 (2025): 14
37. Draine, B., *et al.*, *Detection of Deuterated Hydrocarbon Nanoparticles in the Whirlpool Galaxy, M51*. ApJL 984 (2025): L42
36. Williams, T., *et al.*, *The resolved star-formation efficiency of early-type galaxies*. MNRAS 538 (2025): 3219-3246
35. Pathak, D., *et al.*, *Linking Stellar Populations to H II Regions across Nearby Galaxies. II. Infrared Reprocessed and UV Direct Radiation Pressure in H II Regions*. ApJ 982 (2025): 140

34. Goicoechea, J., *et al.*, *PDRs4All: XII. Far-ultraviolet-driven formation of simple hydrocarbon radicals and their relation with polycyclic aromatic hydrocarbons*. A&A 696 (2025): A100
33. Zannese, M., *et al.*, *PDRs4All: XI. Detection of infrared CH<sup>+</sup> and CH<sub>3</sub><sup>+</sup> rovibrational emission in the Orion Bar and disk d203-506: Evidence of chemical pumping*. A&A 696 (2025): A99
32. Whitmore, B., *et al.*, *Empirical SED Templates for Star Clusters Observed with HST and JWST: No Strong PAH or IR Dust Emission after 5 Myr*. ApJ 982 (2025): 50
31. Dale, D., *et al.*, *PAH Feature Ratios around Stellar Clusters and Associations in 19 Nearby Galaxies*. AJ 169 (2025): 133
30. Chastenet, J., *et al.*, *The Resolved Behavior of Dust Mass, Polycyclic Aromatic Hydrocarbon Fraction, and Radiation Field in ~800 Nearby Galaxies*. ApJS 276 (2025): 2
29. Pingel, N., *et al.*, *The Local Group L-band Survey: The First Measurements of Localized Cold Neutral Medium Properties in the Low-metallicity Dwarf Galaxy NGC 6822*. ApJ 974 (2024): 93
28. Goicoechea, J., *et al.*, *PDRs4All: X. ALMA and JWST detection of neutral carbon in the externally irradiated disk d203-506: Undepleted gas-phase carbon*. A&A 689 (2024): L4
27. Sutter, J., *et al.*, *The Fraction of Dust Mass in the Form of Polycyclic Aromatic Hydrocarbons on 1050 pc Scales in Nearby Galaxies*. ApJ 971 (2024): 178
26. Williams, T., *et al.*, *PHANGS-JWST: Data-processing Pipeline and First Full Public Data Release*. ApJS 273 (2024): 13
25. Mayker Chen, N., *et al.*, *H $\alpha$  Emission and H II Regions at the Locations of Recent Supernovae in Nearby Galaxies*. AJ 168 (2024): 5
24. Fuente, A., *et al.*, *PDRs4All. IX. Sulfur elemental abundance in the Orion Bar*. A&A 687 (2024): A87
23. Van De Putte, D., *et al.*, *PDRs4All. VIII. Mid-infrared emission line inventory of the Orion Bar*. A&A 687 (2024): A86
22. Zannese, M., *et al.*, *OH as a probe of the warm-water cycle in planet-forming disks*. Nature Astronomy 8 (2024): 577-586
21. Schroetter, I., *et al.*, *PDRs4All. VII. The 3.3  $\mu$ m aromatic infrared band as a tracer of physical properties of the interstellar medium in galaxies*. A&A 685 (2024): A78
20. Pasquini, S., *et al.*, *PDRs4All. VI. Probing the photochemical evolution of PAHs in the Orion Bar using machine learning techniques*. A&A 685 (2024): A77
19. Elyajouri, M., *et al.*, *PDRs4All. V. Modelling the dust evolution across the illuminated edge of the Orion Bar*. A&A 685 (2024): A76
18. Peeters, E., *et al.*, *PDRs4All: III. JWST's NIR spectroscopic view of the Orion Bar*. A&A 685 (2024): A74
17. Habart, E., *et al.*, *PDRs4All. II. JWST's NIR and MIR imaging view of the Orion Nebula*. A&A 685 (2024): A73
16. Berné, O., *et al.*, *A far-ultravioletdriven photoevaporation flow observed in a protoplanetary disk*. Science 383 (2024): 988-992

15. Teng, Y., *et al.*, *Star Formation Efficiency in Nearby Galaxies Revealed with a New CO-to-H<sub>2</sub> Conversion Factor Prescription*. ApJ 961 (2024): 42
14. Pathak, D., *et al.*, *A Two-Component Probability Distribution Function Describes the Mid-IR Emission from the Disks of Star-forming Galaxies*. AJ 167 (2024): 39
13. Changala, P., *et al.*, *Astronomical CH<sub>3</sub><sup>+</sup> rovibrational assignments. A combined theoretical and experimental study validating observational findings in the d203-506 UV-irradiated protoplanetary disk*. A&A 680 (2023): A19
12. Berné, O., *et al.*, *Formation of the methyl cation by photochemistry in a protoplanetary disk*. Nature 621 (2023): 56-59
11. Roberts, I., *et al.*, *VERTICO. VI. Cold-gas asymmetries in Virgo cluster galaxies*. A&A 675 (2023): A78
10. Jiménez-Donaire, M., *et al.*, *VERTICO. III. The Kennicutt-Schmidt relation in Virgo cluster galaxies*. A&A 671 (2023): A3
9. Leroy, A., *et al.*, *PHANGS-JWST First Results: A Global and Moderately Resolved View of Mid-infrared and CO Line Emission from Galaxies at the Start of the JWST Era*. ApJL 944 (2023): L10
8. Leroy, A., *et al.*, *PHANGS-JWST First Results: Mid-infrared Emission Traces Both Gas Column Density and Heating at 100 pc Scales*. ApJL 944 (2023): L9
7. **Chown, R.**, *et al.*, *Correction to: The cold gas and dust properties of red star-forming galaxies*. MNRAS 518 (2023): 4536-4536
6. Zabel, N., *et al.*, *Erratum: “VERTICO II: How H I-identified Environmental Mechanisms Affect the Molecular Gas in Cluster Galaxies” (2022, ApJ, 933, 10)*. ApJ 942 (2023): 56
5. Gao, Y., *et al.*, *The Correlation between WISE 12  $\mu$ m Emission and Molecular Gas Tracers on Subkiloparsec Scales in Nearby Star-forming Galaxies*. ApJ 940 (2022): 133
4. Zabel, N., *et al.*, *VERTICO II: How H I-identified Environmental Mechanisms Affect the Molecular Gas in Cluster Galaxies*. ApJ 933 (2022): 10
3. Berné, O., *et al.*, *PDRs4All: A JWST Early Release Science Program on Radiative Feedback from Massive Stars*. PASP 134 (2022): 054301
2. Brown, T., *et al.*, *VERTICO: The Virgo Environment Traced in CO Survey*. ApJS 257 (2021): 21
1. Brown, T., *et al.*, *VERTICO: Virgo Environment Traced In CO*. (2020): 13

### **C. Other papers with significant contributions prior to PhD**

*Summary: 17 papers, 833 citations as of May , 2026.*

17. Sánchez, J., *et al.*, *Mapping gas around massive galaxies: cross-correlation of DES Y3 galaxies and Compton-y maps from SPT and Planck*. MNRAS 522 (2023): 3163-3182
16. Omori, Y., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. I. Construction of CMB lensing maps and modeling choices*. Phys. Rev. D 107 (2023): 023529
15. Abbott, T., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck. III. Combined cosmological constraints*. Phys. Rev. D 107 (2023): 023531

14. Chang, C., *et al.*, *Joint analysis of Dark Energy Survey Year 3 data and CMB lensing from SPT and Planck . II. Cross-correlation measurements and cosmological constraints.* Phys. Rev. D 107 (2023): 023530
13. Anbajagane, D., *et al.*, *Shocks in the stacked Sunyaev-Zel'dovich profiles of clusters II: Measurements from SPT-SZ + Planck Compton-y map.* MNRAS 514 (2022): 1645-1663
12. Salvati, L., *et al.*, *Combining Planck and SPT Cluster Catalogs: Cosmological Analysis and Impact on the Planck Scaling Relation Calibration.* ApJ 934 (2022): 129
11. Bleem, L., *et al.*, *CMB/kSZ and Compton-y Maps from 2500 deg<sup>2</sup> of SPT-SZ and Planck Survey Data.* ApJS 258 (2022): 36
10. Omori, Y., *et al.*, *Dark Energy Survey Year 1 Results: Cross-correlation between Dark Energy Survey Y1 galaxy weak lensing and South Pole Telescope+Planck CMB weak lensing.* Phys. Rev. D 100 (2019): 043517
9. Omori, Y., *et al.*, *Dark Energy Survey Year 1 Results: Tomographic cross-correlations between Dark Energy Survey galaxies and CMB lensing from South Pole Telescope +Planck.* Phys. Rev. D 100 (2019): 043501
8. Prat, J., *et al.*, *Cosmological lensing ratios with DES Y1, SPT, and Planck.* MNRAS 487 (2019): 1363-1379
7. Mocanu, L., *et al.*, *Consistency of cosmic microwave background temperature measurements in three frequency bands in the 2500-square-degree SPT-SZ survey.* JCAP 2019 (2019): 038
6. Abbott, T., *et al.*, *Dark Energy Survey year 1 results: Joint analysis of galaxy clustering, galaxy lensing, and CMB lensing two-point functions.* Phys. Rev. D 100 (2019): 023541
5. Simard, G., *et al.*, *Constraints on Cosmological Parameters from the Angular Power Spectrum of a Combined 2500 deg<sup>2</sup> SPT-SZ and Planck Gravitational Lensing Map.* ApJ 860 (2018): 137
4. Hou, Z., *et al.*, *A Comparison of Maps and Power Spectra Determined from South Pole Telescope and Planck Data.* ApJ 853 (2018): 3
3. Aylor, K., *et al.*, *A Comparison of Cosmological Parameters Determined from CMB Temperature Power Spectra from the South Pole Telescope and the Planck Satellite.* ApJ 850 (2017): 101
2. Baxter, E., *et al.*, *Joint measurement of lensing-galaxy correlations using SPT and DES SV data.* MNRAS 461 (2016): 4099-4114
1. Soergel, B., *et al.*, *Detection of the kinematic Sunyaev-Zel'dovich effect with DES Year 1 and SPT.* MNRAS 461 (2016): 3172-3193

#### D. Abstracts in proceedings

4. **R. Chown**, et al. "Polycyclic Aromatic Hydrocarbon and CO (2-1) Emission at 50-150 pc Scales in 70 Nearby Galaxies." AstroPAH Newsletter, 117 (April 2025).
3. **R. Chown**, et al. "PDRs4All IV. An embarrassment of riches: the Aromatic Infrared Bands in the Orion Bar." AstroPAH Newsletter, 104 (December 2023).
2. **R. Chown**, et al. "The Relationship Between Mid-infrared and CO Emission at kpc Scales in Nearby Galaxies." Bulletin of the American Astronomical Society, Vol. 53, No. 6 (2021).
1. T. Brown, et al. (incl. **R. Chown**) "VERTICO: Virgo Environment Traced In CO." Scientific Meeting of the Spanish Astronomical Society, 13-15 July 2020.

## PUBLIC TALKS

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- |                                                                             |               |
|-----------------------------------------------------------------------------|---------------|
| 5. Public Astro Night, Queen's University                                   | Kingston, ON  |
| Public talk on star formation and galaxies with JWST                        | 8 June 2024   |
| 4. Amica Retirement Home                                                    | London, ON    |
| “Are We Alone?” – a talk about searching for life in the Universe with JWST | June 2023     |
| 3. Parkwood Institute                                                       | London, ON    |
| “Are We Alone?” – a talk about searching for life in the Universe with JWST | January 2023  |
| 2. JWebbinar 23: PDRs4All Community Telecon                                 |               |
| Gave a public talk on status of JWST observations from PDRs4All             | December 2022 |
| 1. Astronomy on Tap                                                         | Montreal, QC  |
| “Patterns in the Cosmic Microwave Background”                               | May 2017      |

## OUTREACH

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|---------------------------------------------------------------------------------------------------------------------------|----------------------|
| Canadian Bushplane Heritage Centre                                                                                        | Sault Ste. Marie, ON |
| Facilitated 2-day “Health in Space” hands-on event for Grade 6 students                                                   | March 2026           |
| CBC Coverage: <a href="https://www.cbc.ca/player/play/video/9.7149669">https://www.cbc.ca/player/play/video/9.7149669</a> |                      |
| Algoma University, Global Game Jam                                                                                        | Sault Ste. Marie, ON |
| Co-organizer of the Algoma University Global Game Jam                                                                     | 2026                 |
| STEAM Factory, Ohio State University                                                                                      | Columbus, OH         |
| Gave a talk on “stellar birth”                                                                                            | 2024                 |
| McMaster Undergraduate Physics Society                                                                                    | Hamilton, ON         |
| Organized and gave a seminar on Python for undergraduates in physics at McMaster                                          | 2021                 |
| Astro McGill                                                                                                              | McGill University    |
| Helped facilitate public Astro Nights                                                                                     | 2016-2017            |
| Physics Matters                                                                                                           | McGill University    |
| Helped organize this series of public talks on physics                                                                    | 2016-2017            |
| Eureka! Science Festival                                                                                                  | Montreal, QC         |
| Instructed physics experiments for elementary school students                                                             | 2015                 |

## INVITED SEMINARS

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|--------------------------------------------------------------------------------|---------|
| 9. Astro Tea talk, Indiana University, Bloomington                             | 01/2025 |
| 8. Astronomy Journal Club, Waterloo Center for Astrophysics                    | 10/2024 |
| 7. Astronomy Seminar, Queen's University Department of Physics & Astronomy     | 06/2024 |
| 6. PHANGS Colloquium                                                           | 02/2024 |
| 5. Astronomy Journal Club, McMaster University                                 | 11/2023 |
| 4. CANadian Virtual Astronomy Seminar (CANVAS)                                 | 03/2023 |
| 3. Ringberg Seminar Series, MPIA, Germany                                      | 05/2021 |
| 2. Extragalactic Database for Galaxy Evolution Meeting, University of Maryland | 05/2021 |
| 1. Shanghai Astronomical Observatory, Shanghai, China                          | 08/2018 |

## CONTRIBUTED TALKS

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|-------------------------------------------------------------------------------------------------|---------|
| <i>Star Formation, Stellar Feedback, and the Ecology of Galaxies</i> , Visegrad, Hungary        | 05/2025 |
| <i>PHANGS Collaboration Meeting</i> , Nice, France                                              | 02/2025 |
| <i>The Physics and Impact of Astrophysical Dust: Star Formation to Cosmology</i> , Aspen, CO    | 03/2024 |
| <i>PHANGS Collaboration Meeting</i> , Garching, Germany                                         | 02/2024 |
| <i>Illuminating the Dusty Universe: A Tribute to the Work of Bruce Draine</i> , Florence, Italy | 10/2023 |
| <i>Symposium on the Life Cycle of Cosmic PAHs</i> , Aarhus University, Denmark                  | 09/2022 |
| <i>KIAA Forum on Gas in Galaxies for Early Career Scientists</i> , Peking University, Beijing   | 11/2021 |
| <i>238th AAS Meeting – Dissertation Talk</i>                                                    | 7/2021  |
| <i>Exploring Gas in and Around Galaxies meeting</i> , Tsinghua University, Beijing              | 7/2018  |
| <i>KIAA Forum on Gas in Galaxies</i> , Peking University, Beijing                               | 6/2018  |
| <i>SDSS Chinese MaNGA Meeting</i> , University of Chinese Academy of Sciences, Beijing          | 6/2018  |
| <i>South Pole Telescope Collaboration Conference</i> , University of Chicago, IL                | 7/2017  |
| <i>The Centre for Research in Astrophysics of Quebec (CRAQ) Annual Meeting</i> , Montreal, QC   | 5/2017  |
| <i>South Pole Telescope Collaboration Conference</i> , University of Chicago, IL                | 8/2016  |
| <i>South Pole Telescope Collaboration Conference</i> , University of Chicago, IL                | 7/2015  |

## CONFERENCE POSTERS

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|                                                                                          |        |
|------------------------------------------------------------------------------------------|--------|
| <i>First Science Results from JWST</i> , STScI, Baltimore                                | 9/2023 |
| <i>Canadian Astronomical Society Annual Conference</i> , York University, Toronto, ON    | 5/2020 |
| <i>Views on the ISM in galaxies in the ALMA era</i> , University of Bologna, Italy       | 9/2019 |
| <i>Canadian Astronomical Society Annual Conference</i> , McGill University, Montreal, QC | 6/2019 |

## APPROVED OBSERVING PROPOSALS

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### As P.I. or Co-P.I.:

5. VLA Cycle 26A (co-PI; PI: Eibensteiner, C.) – “High Quality HI Coverage of the Northern PHANGS JWST+HST+ALMA Targets.”
4. ALMA Cycle 11 (2024.1.01179.S; P.I.) – “A Complete View of Low Metallicity Star Forming Complexes in the Local Group Dwarf NGC 6822.”
3. JCMT 21A (P.I.) – “Measuring global CO(2-1) to supplement interferometric observations from the EDGE survey.”
2. JCMT 18B (P.I.) – “Observing red star-forming galaxies from xCOLD GASS with SCUBA-2.”
1. JCMT 18A (P.I.) – “Extending the JINGLE RxA Samples to Include ‘Red Misfit’ Galaxies.”

### As Co-I.:

18. JWST Cycle 5: “Where the PAH-CO Relation Breaks: JWST Spectroscopy of High CO/PAH Ridges in NGC 1566.” PI: J. Kim.
17. JWST Cycle 5: “Dust in the Deserts: Understanding PAH Emission in Quiescent Bulges and Star-formation Deserts.” PI: D. Pathak.
16. JWST Cycle 5: “Resolving Stellar Feedback in Action via NIRCcam and MIRI Imaging of Very Nearby Galaxies.” PI: A. Leroy.

15. JWST Cycle 5: “A Complete Mapping of Stellar Evolution and the Interstellar Medium across M33.” PI: E. Rosolowsky.
14. JWST Cycle 4: “Exploring the Last Watering Hole of Low Metallicity PAH Emission: Deep Spectroscopic Observations of Sextans A.” PI: L. Tarantino.
13. JWST Cycle 4: “Resolving the thin disk of a nearby Milky Way analog.” PI: A. Leroy.
12. ALMA Cycle 11: “Linking Molecular Cloud Structure to Massive Star Formation: 5000 molecular clouds, filaments, and bubbles across M33.” PI: E. Koch.
11. ALMA Cycle 11: “Completing the Virgo High-resolution CO 2-1 Survey: Dissecting Galaxy Quenching with Molecular Cloud Scale Micro-physics”. PI: J. Sun.
10. ALMA Cycle 11: “The Last Refuge of CO and PAH Emission in the Most Metal Poor Galaxies.” PI E. Tarantino.
9. HST Cycle 32: “Decoding stellar feedback in action with an HST+MUSE+JWST full-disk survey of starburst galaxy prototype NGC 253.” PI D. Thilker.
8. HST Cycle 32: “Bringing HST to the VLA: The Interaction of Stars and Gas in the Local Group.” PI J. Dalcanton.
7. ALMA Cycle 10: “Virgo High-resolution CO(2-1) Survey: Dissecting Galaxy Quenching with Molecular Cloud Scale ‘Micro-physics’”. PI: J. Sun.
6. JWST Cycle 3: “A Systematic Study of the 3.3 - 3.5  $\mu\text{m}$  PAH Features at  $z = 0$  with Archival NIRSpec Observations.” PI K. Sandstrom.
5. ALMA Cycle 9 – “The chemical richness of the Orion Bar and its role as a lab of  $\text{CH}_3\text{CN}$  chemistry in disk-evolved systems.” P.I. F. Alarcón.
4. ALMA Cycle 7 (Large Program) – “The Virgo Environment Traced in CO survey (VERTICO).” P.I. T. Brown.
3. ALMA Cycle 7 – “Mapping CO emission in galaxies from the JINGLE survey.” P.I. C.D. Wilson.
2. JCMT 20A (Large Program) – “JINGLE at the edge: the ISM of starbursts and green valley galaxies.” P.I. L.-H. Lin.
1. JCMT 19B – “Observing CO(2-1) in Red Star-forming Galaxies.” P.I. L.-H. Lin.

## COLLABORATIONS

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6. The Physics at High Angular Resolution in Nearby Galaxies (PHANGS) Collaboration<br><a href="http://www.phangs.org">http://www.phangs.org</a>                                                                      | 10/2023–  |
| 5. The Local Group L-Band Survey Collaboration<br><a href="https://www.lglbs.org">https://www.lglbs.org</a>                                                                                                           | 10/2023–  |
| 4. PDRs4All: Radiative Feedback from Massive Stars as Traced by Multiband Imaging and Spectroscopic Mosaics – A JWST Early Release Science (ERS) Program<br><a href="https://pdrs4all.org">https://pdrs4all.org</a>   | 10/2021–  |
| 3. The Virgo Environment Traced in CO survey (VERTICO; an ALMA Large Program)<br><a href="https://sites.google.com/view/verticosurvey/home">https://sites.google.com/view/verticosurvey/home</a>                      | 2019–2022 |
| 2. The JCMT Dust & Gas in Nearby Galaxies Legacy Survey (JINGLE)<br><a href="https://www.eaobservatory.org/jcmt/science/large-programs/jingle/">https://www.eaobservatory.org/jcmt/science/large-programs/jingle/</a> | 2017–2021 |

1. The South Pole Telescope Collaboration 2013–2017  
<https://pole.uchicago.edu/public/Home.html>

## **AWARDS**

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|--------------------------------------------------------------|---------------|
| Visiting Scholar, Tsinghua University, Beijing, China        | 2019          |
| Mitacs Globalink Research Award                              | 2018          |
| McMaster Graduate Fellowship                                 | 2018          |
| McGill University Graduate Excellence Award in Physics       | 2015 and 2016 |
| Carl Reinhardt Fellowship                                    | 2015          |
| McGill and Novelis Global Technologies Summer Research Award | 2014          |
| McGill Summer Research Award                                 | 2013 and 2015 |